QUIN PRO (Decathlon) Mobile App Integration Guide

1. Feature summary

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2. Advertising

2.1 When advertising starts and stops

Behaviour	Rule
Not advertising at boot	Advertising is **not** started at power-up or wake.
Not advertising after disconnect	The SDK auto-restart is disabled. After a disconnect the helmet does **not** advertise.
Start trigger 1: double tap	Exactly 2 taps detected, helmet **not connected** and **not already advertising**.
Start trigger 2: offline crash saved	After a crash is stored to flash while disconnected, advertising is started so the app can fetch it.
Start trigger 3: emergency interrupt	Internal emergency event (`EmergEventDetect`), if that interrupt is enabled, while disconnected.
Stop: timeout	Advertising stops automatically after **20 s** (current firmware: `APP_ADV_TIMEOUT_IN_SECONDS = 20`).
Stop: connection	Advertising stops when a central connects.
Stop: crash in progress	If a wake/crash event starts while advertising and the device is not connected, advertising is stopped until the data is saved to flash.
Not started when	Already connected, or already advertising.

Double-tap detection: individual taps closer than about 175 ms apart are ignored. The tap count is evaluated 600 ms after the last accepted tap. 2 taps = double tap. While **connected**, a double tap instead triggers a forced battery read and a battery status notification (section 8.2).

2.2 Advertising packet layout

The advertising data contains, in order: **Flags**, **Manufacturer Specific Data**, **Name**. Legacy advertising is limited to 31 bytes, so the name is encoded last and truncated if it does not fit.

AD structure	Bytes	Content
Flags	02 01 05	LE Limited Discoverable + BR/EDR not supported
Manufacturer Specific Data	16 bytes total	See table below

AD structure	Bytes	Content
Local Name	up to 12 bytes	See note on truncation

Manufacturer Specific Data (AD type 0xFF), payload after the length and type bytes:

Offset	Size	Value	Meaning
0–1	2	D6 0E	Company identifier 0x0ED6 (little-endian on air)
2	1	MAC[5]	MAC address, **most significant byte first**
3	1	MAC[4]	
4	1	MAC[3]	
5	1	MAC[2]	
6	1	MAC[1]	
7	1	MAC[0]	
8	1	0x08	Decathlon
9	1	0x08	Snowboarding
10	1	0x04	Helmet (nRF5 SDK)
11	1	0xB3	Flash storage enabled, 1000 Hz, 400 g
12	1	0–100	**Battery percentage** (second-last byte of the manufacturer payload)
13	1	0–255	**Stored crash/offline record count** (last byte), clamped to 255

Total manufacturer payload after the company ID is 12 bytes; including the company ID it is 14 bytes (AD length byte = 15).

Name truncation: with Flags (3 bytes) and this manufacturer structure (16 bytes), 12 bytes remain for the name AD structure, which holds at most 10 name characters. The default name QuinPro0808 has 11 characters, so it is expected to be advertised as a **Shortened Local Name** (QuinPro080). The full name is available from the GAP Device Name after connecting. **Do not rely**

on the advertised name to identify the device. Identify it by company ID `0x0ED6` plus the bytes `08 08 04 B3`.

2.3 Scan response

Complete list of service UUIDs: the 128-bit data service `8925D23D-...` (`0xD23D`) and the 16-bit Device Information service `0x180A`.

2.4 Reading battery and record count without connecting

Parse the manufacturer data in the scan result:

```
payload = manufacturer_data (after company ID 0x0ED6)
battery_percent = payload[10]
record_count    = payload[11]
```

`record_count` is the number of crash records in the offline data partition at the time advertising started (data not yet erased). If the count is greater than 0 the app can connect and request the offline data (section 3.3).

3. Commands and responses

3.1 Command basics

Only the first byte of a write to 0x6166, 0x6266 and 0x1002 is used as the command value, except the 4-byte colour command on 0x1002 (section 3.4).

3.2 Required connection sequence

1. Connect.
2. Request MTU ≥ 129 (recommended 185 or more).
3. Discover services and characteristics.
4. **Enable notifications on `0x6166`, `0x6266` (if live mode is used) and `0x1002`.**
5. Write the RTC.
6. Optionally read battery (section 8.2) and request offline data.

3.3 Data-mode characteristic `0x6166`: commands

Value written	Meaning	Device behaviour	Response
`0x01`	**Send offline crash data**	If records exist, starts sending after about 2 s (section 7.4). Records already downloaded are skipped.	Crash packets (126 bytes each) on `0x6166`, then end marker. If none: end marker immediately.
`0x02`	**Delete offline crash data**	Walks all records without sending them, then **erases the offline partition**. Also	End marker `01 33 55 AA` when done.

Value written	Meaning	Device behaviour	Response
		clears the red crash LED and crash status.	
0x03	**Send battery/temperature log**	Replays all log records (section 8.3).	Log chunks on 0x6166, then end marker.
0x04	**Delete battery/temperature log**	Skips through all records, then erases the log partition.	End marker when done.

Values 0x03 and 0x04 exist only because battery logging is compiled in (BATT_LOG_ENABLE = 1).

End marker: a 4-byte notification on 0x6166: 01 33 55 AA. It signals the end of an offline-data or battery-log transfer (including "nothing to send"). Crash data packets are 126 bytes and log chunks are multiples of 10 bytes, so a 4-byte notification is unambiguous.

Warning: enable notifications first. On 0x6166 the firmware does not check that notifications are enabled. If the app writes 0x01 before enabling the CCCD, notifications are silently dropped but the records are still **marked as transmitted** and will not be re-sent. Always enable notifications before sending 0x01.

Warning: one transfer at a time. Do not start a second request, and avoid disconnecting, mid-transfer. The offline replay does not appear to reset its state on disconnect, so behaviour after a mid-transfer drop (resume, or the helmet being held awake) should be verified on hardware.

Timing of a transfer (approximate):

Item	Value
Delay between request and first packet	~2 s (offline), ~20 ms (battery log)
Packet spacing, offline replay	120 ms
Packet spacing, live (online) crash send	250 ms
Gap between records	~200 ms
One crash record	64 packets ≈ 8 s

3.4 Alert characteristic 0x1002: commands

Write a single byte. Both the numeric value and the ASCII digit are accepted.

Value(s)	Command	Device action	Response (notification on 0x1002)
0x01 or '1' (49)	**Acknowledge crash/SOS**	Turns off the red crash LED, clears the major-crash, SOS and crash-detect flags, and clears any BLE-set colour (testing only)	none
0x02 or '2' (50)	Reserved	No action	none
0x03 or '3' (51)	**Start gyro calibration** (factory)	Blinks blue; starts calibration and live capture. Offsets are stored to flash when calibration ends.	Calibration Started, later Calibration Stopped, then gx,gy,gz offsets
0x04 or '4' (52)	**Reset calibration to zero** (factory)	Clears stored gyro offsets	none
0x05 or '5' (53)	Reset accelerometer high-pass filter	Requests HP filter reset	none
0x06 or '6' (54)	Reset pedometer count	Resets step counter	none
0x07 or '7' (55)	**Capture base orientation** (factory)	Averages accelerometer samples to set the default orientation	Avg Res:x.xx, then Orient updated
0x08 or '8' (56)	Read raw accel/gyro values	Reads one sample	Acc: x.xx,y.xx,z.xx or Ax ..., Ay ..., Az ..., p ..., r ...
0x09 or '9' (57)	**Reset IMU timestamp**	Zeros the IMU hardware timestamp, overflow count and stored state	none
0x0A, 'A' or 'a'	**Sync / get IMU timestamp**	Saves current timestamp state to flash and reports it	TS:XXXXXXXX (section 6.1)
0x0B + R, G, B (4 bytes)	**Set LED colour (testing only)**	Shows the colour on the LED. Each channel is 0 to 100,	RGB,r,g,b (or RGB,CHG if charging)

Value(s)	Command	Device action	Response (notification on 0x1002)
		and values above 100 are clamped. 0,0,0 clears it. Ignored while charging.	
anything else	Ignored		

Commands 3–8 are factory/diagnostic. They are not needed for normal app operation. The helmet only sends calibration text when not on the charger.

The colour command is the one exception to the single-byte rule: it must be exactly 4 bytes (0x0B, R, G, B). A write starting with 0x0B of any other length is not treated as a colour command.

4. Activity, inactivity, sleep and wake

4.1 Timing summary

Parameter	Value	Source
Motion threshold	1.5 g (high-pass filtered)	`WAKE_UP_THS` = 0x06 × 0.25 g
Inactivity timeout	about **10 s** of no motion above threshold (≈9.85 s; `SLEEP_DUR` = 2 at 104 Hz)	LSM6DS3 hardware
Sleep timeout	**50 s** after the inactivity event with no activity in between (total about 60 s after last motion)	`rst_check_inactivity()`
Advertising timeout	20 s	Section 2

4.2 States

State	Entered when	Gyro	Notification	LED heartbeat
Active	Boot with motion, or motion above threshold after inactivity	On	`ACT` (only if previous state was inactive)	Every ~1 s (100 ms on / 900 ms off)
Inactive	~10 s with no motion above threshold	**Powered down by hardware**	`INACT`	Every ~5 s (100 ms on / 4900 ms off)

State	Entered when	Gyro	Notification	LED heartbeat
Sleep (system off)	50 s after `INACT` with no motion, no transfer in progress, not advertising, not charging	Off	none (link is not available)	Off

4.3 Notifications

`INACT` and `ACT` (see the states table) are sent on `0x1002` as ASCII text, only while connected. They are not queued if disconnected.

4.4 Sleep conditions

The helmet enters system-off sleep only when **all** of the following hold when the 50 s timer expires:

- no offline-data transfer, offline delete, or battery-log transfer/delete in progress
- no online (live) crash packets still being sent
- no flash read/erase busy
- not advertising
- not on the charger

If any is not true, the timer is re-armed for another 50 s.

A connection does not by itself keep the helmet awake: if the helmet is still and the BLE link is open, it will still go to sleep after the same timers, which drops the link.

4.5 Wake

Wake source	Result
Shake / motion above 1.5 g	Wakes from sleep. Firmware reboots. If the IMU reports a wake-up event at boot the helmet starts Active (LED heartbeat 900 ms off-time); otherwise it starts Inactive.

Wake from sleep is a reset: advertising is **not** started and the BLE link is not present. The user must double-tap to advertise (section 2).

4.6 Gyro off while inactive (feature 7)

While inactive the IMU gyro is powered down by hardware. If a crash occurs while the helmet is inactive, the first samples of the crash window can contain **empty gyro data**. The firmware detects samples taken during the gyro power-down and turn-on settle window (about 100 ms) and **blanks gyro fields to 0** in those samples. Accelerometer data is always valid.

App guidance: treat `Gx = Gy = Gz = 0` for the earliest samples of a crash window as "not measured", not as real zero rotation.

5. Data modes

5.1 Crash data (primary mode)

Crash events are captured by the firmware and delivered as a fixed sequence of 126-byte packets (section 7). They are delivered either:

- **Online**: sent live over `0x6166` immediately after the event (if connected and the configuration allows), or
- **Offline**: stored to flash while disconnected and sent when the app writes `0x01` to `0x6166`.

5.2 Live mode on `0x6266` (diagnostic)

Write to `0x6266`	Effect
`0x01`	Live mode on (requires notifications enabled on `0x6266`)
`0x00`	Live mode off

Live-mode payload structure (170 bytes): `PacketType` (1) = `0x22`, `AlertType` (1), frame number (2), timestamp (4), then 9 samples × 18 bytes (`Gx Gy Gz Ax Ay Az Ix Iy Iz`, each int16). This is a legacy/diagnostic mode: in this firmware build the live-mode send routine is not called from the main loop, and live IMU capture is only exercised during factory calibration. It is not part of the Decathlon feature set. Live mode is reset to off on disconnect.

6. Timestamps

6.1 IMU hardware timestamp (alert-characteristic commands)

The IMU keeps a free-running 24-bit counter at 25 µs per tick. The firmware extends it with an overflow count and persists it to flash at sleep, on sync, and at boot.

Command (written to `0x1002`)	Effect
`0x09` / `'9'`	Reset: zero the IMU timestamp, overflow count and stored copy
`0x0A` / `'A'` / `'a'`	Sync: store current state, then notify the value

Response format (notification on `0x1002`, ASCII, 11 characters):

```
TS:XXXXXXXX
```

XXXXXXXX is an 8-digit upper-case hexadecimal number:

```
extended_ts = (overflow_count << 24) | timestamp_24bit
elapsed_seconds ≈ extended_ts × 25e-6
```

Example: `TS:00012345` ➜ 0x00012345 = 74565 ticks ➜ about 1.86 s.

Because the overflow interrupt fires near `0xFF0000` rather than at `0x1000000`, each overflow can under-count by up to 65,535 ticks (1.6 s), so the result is an approximation over long periods.

Typical use: send `0x09` once at manufacturing/first use, then send `0x0A` on each connection and compare against the app clock to map IMU time to wall-clock time.

7. Crash data packet format

7.1 Packet (online and offline are identical)

Each packet is **126 bytes**, sent as a notification on `0x6166`.

Byte	Size	Field	Description
0	1	Frame number	1-based sequence number within one crash record (1–64)
1	1	Sensor type	`0x01` = IMU data (pre **and** post, see 7.3), `0x02` = high-g impact data
2	1	Packet type	See 7.2. `0x11` for online (live) packets; offline packets carry the stored code
3–5	3	**Crash ID**	Unique ID of this crash event, the same in all 64 packets of the event
6–125	120	Samples	See 7.5

Crash ID (bytes 3–5): a 3-byte pseudo-random value generated once when the crash is first detected and reused for both the live stream and the flash copy. A crash that is streamed live and also stored will carry the same ID, so the app can de-duplicate. The generator is seeded at boot from the stored IMU timestamp and battery voltage, so it differs across power cycles. Treat it as an opaque 24-bit identifier (collisions are possible but unlikely).

7.2 Packet type (byte 2)

Value	Meaning	Where used
`0x11`	Online / live crash packet	Online stream
`0x22`	Manual live packet	Live mode (5.2)
`0x33`	ADL (activity of daily living)	Reserved

Value	Meaning	Where used
0x43	Fall, stored offline	Offline replay
0x53	Crash after free-fall (fall from height), stored offline	Offline replay
0x63	Impact, stored offline	Offline replay
0x73, 0x83, 0x93	Low-power variants of fall / crash-FF / impact	Defined but not used in this firmware

7.3 Frame sequence of one crash record

Frames	Sensor type (byte 1)	Content	Samples	Rate	Samples per packet
1–30	0x01	**Pre-event IMU** buffer	300	104 Hz (≈2.9 s)	10
31–60	0x01	**Post-event IMU** buffer	300	52 Hz (≈5.8 s)	10
61–64	0x02	**High-g impact** buffer	80	about 1 kHz (≈80 ms window)	20

Pre and post IMU packets both carry sensor type 0x01. **Distinguish them by frame number** (1–30 pre, 31–60 post). In the online stream the post-event block is sent as an internal type 0x04 but byte 1 is rewritten to 0x01. The frame counter is not reset between blocks.

The frame counter and packet sequence restart at 1 for each new crash record. Packet spacing is given in section 3.3. A complete record is 64 × 126 = 8,064 bytes.

7.4 Offline record in flash (reference)

One offline record is 7,692 bytes in flash. It holds the same information as the 64 packets:

Field	Size	Notes
Data flag	1	0xFE valid, 0xF8 already transmitted, 0xFC invalid, 0xFF empty
Packet type	1	Section 7.2
Alert type	1	1 major, 2 minor, 3 brake, 4 manual
Frame count	2	

Field	Size	Notes
Timestamp	4	IMU clock when stored
Crash ID	3	
Pre-event IMU	3,600	300 × 12 bytes
Post-event IMU	3,600	300 × 12 bytes
High-g	480	80 × 6 bytes

The partition is roughly 1 MB, so on the order of 130 records fit; the firmware has circular-overwrite logic for when it is full. The record count in the advertisement is clamped to 255.

7.5 Sample layouts

IMU sample (sensor type `0x01`), 12 bytes, little-endian int16, 10 per packet:

Offset	Field	Scale
0	Gx	0.07 °/s per LSB (±2000 °/s)
2	Gy	0.07 °/s per LSB
4	Gz	0.07 °/s per LSB
6	Ax	0.488 mg per LSB (±16 g) ➜ g = raw × 0.000488
8	Ay	0.488 mg per LSB
10	Az	0.488 mg per LSB

The firmware already inverts the X axes (Gx and Ax) and subtracts the stored gyro calibration offset from Gx. Use values as delivered. Early samples may have gyro fields equal to 0 (section 4.6).

High-g sample (sensor type `0x02`), 6 bytes, 20 per packet:

Offset	Field	Scale
0	Ix	0.195 g per LSB (±400 g)
2	Iy	0.195 g per LSB
4	Iz	0.195 g per LSB

The high-g values are 12-bit two's-complement values stored in a 16-bit field without sign extension (range 0–4095). The app must sign-extend:

```
v = raw & 0x0FFF
if v >= 0x0800: v -= 0x1000
g = v * 0.195
```
Use the Ix / Iy / Iz labels as delivered.

7.6 Reassembly guidance for the app

1. Start a new record when frame number = 1 (or when the crash ID changes).
2. Collect frames 1–64, keyed by (crash ID, frame number); ignore duplicates.
3. Frames 1–30: pre IMU (300 samples); 31–60: post IMU (300 samples); 61–64: high-g (80 samples).
4. A record is complete when 64 distinct frames are received.
5. After the last record, an end marker (01 33 55 AA) arrives on offline transfers. Online crash streams do not send an end marker.
6. If the link is lost, discard the incomplete record and request the data again after reconnecting.

8. Alert (text) notifications on 0x1002

All notifications on 0x1002 are ASCII text with no terminator. The app should split on the leading token. Only sent while connected and with the CCCD enabled.

8.1 Catalogue

Text (examples)	When	Notes
INACT / ACT	Inactivity / activity change	Section 4
TAP:n	Tap sequence finished (600 ms after last tap)	n = tap count. TAP:2 = double tap. n ≥ 7 also sends Noise
Wake	Wake trigger (single-stage)	Start of a possible crash capture
WAKE:tick, ST2-WAKE:tick, ST2-FALL:tick	Wake / free-fall trigger	tick = firmware millisecond tick
ST1-t:tick	Returned to stage 1 after 5 s without a crash	
FreeFall:tick	Free-fall detected	
C:tick	Major crash interrupt	
MINOR CRASH	Minor impact interrupt	
Impact, Fall, CRASH-FF	**Classification** of the finished event	CRASH-FF = fall from height

Text (examples)	When	Notes
ires:x.xx	Resultant impact (g), sent right after the classification	
H:h,start,end	Free-fall height estimate	
Noise	7 or more taps	
Temp:t0,t1,...,t9	Ten temperature readings (°C)	See 8.4
FULL BATTERY,pp,t / MEDIUM BATTERY,pp,t / LOW BATTERY,pp,t	Battery status	See 8.2
TS:XXXXXXXX	Reply to timestamp sync	Section 6.1
Calibration Started / Calibration Stopped / gx,gy,gz	Factory calibration	
Avg Res:x.xx, Orient updated, Acc: ..., Ax ...	Replies to commands 7 and 8	
RGB,r,g,b / RGB,CHG	Reply to the colour command (testing only)	Reply to the colour command. RGB,CHG means the command was ignored because the helmet is charging.

Timing strings (tick) are firmware debug values; do not use them for wall-clock time. Some of the wake / stage strings (Wake, WAKE:, ST1-, ST2-) come from older state logic and may not appear in the current build.

The app should treat unknown alert strings as informational. For crash handling, the classification (Impact, Fall, CRASH-FF) arrives before the data packets. Per the firmware, a plain "ADL" result (no impact) is suppressed from online sending.

8.2 Battery

A battery notification is sent on 0x1002 when connected and either (a) the battery percentage changed since the last reading, or (b) a forced read happens: about 5 s after connecting, and on a double tap while connected.

Format: <STATE>,<percent>,<temperature>

STATE	Percent range
FULL BATTERY	≥ 90
MEDIUM BATTERY	31–89

STATE	Percent range
LOW BATTERY	≤ 30

Example: MEDIUM BATTERY,67,24.500000 (temperature printed as a float).

Reading the alert characteristic 0x1002 returns the last battery string. The percentage is derived from a battery-voltage lookup table with two tables (normal and cold) selected by temperature with 5 °C hysteresis (cold ≤ −5 °C, normal ≥ +5 °C). The battery percentage reported never rises while not charging.

The battery percentage is updated after 20 ADC samples (about 15 s); it is also present in the advertising packet (section 2.2).

8.3 Battery / temperature log

The helmet logs one record every ~15 s regardless of connection. Command 0x03 (section 3.3) replays all records on 0x6166 and then sends the end marker.

Each record is **10 bytes** (packed structure). Each notification carries as many whole records as the negotiated MTU permits, up to 13 per notification (130 bytes). Split each notification into 10-byte records:

Offset	Size	Field
0	1	Data flag (0xFE valid)
1–4	4	Timestamp (device clock)
5	1	Battery percentage
6–7	2	Average battery voltage, mV
8–9	2	Temperature × 10, signed (°C × 10)

Notification spacing is 120 ms. Command 0x04 deletes the log without sending it.

8.4 Temperature

The IMU temperature sensor is read on connect, every 60 s while connected, and about every 15 s by the battery/temperature logger. Every tenth reading the helmet sends one Temp: notification with the last ten values (°C, two decimals), so the cadence is irregular (roughly every 2 minutes while connected).

9. Configuration

9.1 Configuration characteristic 0x1001

Fixed 10-byte structure. Read returns the current values; write replaces them all.

Offset	Field	Default	Meaning
0	enable_low_power	0x00	Low-power flag (currently affects LED colour only)
1	MajorThresh	80	Major crash threshold (g)
2	MinorThresh	20	Minor crash threshold (g)
3	wakedur	0x20	Wake duration setting
4	freefall	0xFB	Free-fall setting
5	minordur	2	Minor crash duration
6	wakethr	0x06	Wake threshold (0.25 g per LSB; 0x06 = 1.5 g)
7	majordur	5	Major crash duration
8	secondthr	40	Second-stage threshold
9	crash_mode	0x01	Offline/online handling (9.2)

9.2 crash_mode: offline / online packets

Bit	Name	Meaning
0 (0x01)	Offline	Store crash data to flash
1 (0x02)	Online	Send crash data live when connected

crash_mode	Offline	Online	Behaviour after a crash is classified
0x00	off	off	Nothing is stored or sent.
0x01 (**default**)	on	off	**Always stored to flash**, even while connected and regardless of

crash_mode	Offline	Online	Behaviour after a crash is classified
			classification. Advertising is started if disconnected. The app fetches with command 0x01.
0x02	off	on	Sent live **only if connected** and the event is classified as significant (impact / fall / crash-FF). Otherwise discarded. Never stored.
0x03	on	on	If connected and significant: sent live (not stored). If connected and not significant: discarded. If **not connected**: stored to flash.

"Significant" excludes events classified as ADL (low-energy impact without fall height or orientation change).

9.3 Writing the configuration

- Write exactly **10 bytes** to 0x1001. Any other length is rejected, except the two special single-byte writes below.
- Single byte 0x01: **reset configuration to defaults** (name unchanged).
- Single byte 0x00: sets the device mode to factory mode (internal use; do not send from the app).
- Writing a valid configuration stores it to flash and then **resets the device after about 100 ms**. The connection will drop. Wait, then reconnect (the helmet will not advertise on its own; the user must double-tap or shake and double-tap).

For safety, always read 0x1001 first, change only the required byte (usually crash_mode), and write back all 10 bytes.

9.4 Device name 0x1004

- Write a name up to 21 characters (the stored field is 21 bytes, so use up to 20 characters to keep a terminator). Longer writes are rejected.
- Single byte 0x01 or '1' (49): reset the name to QuinPro0808.
- Like the configuration write, a name change is stored to flash and the device resets.

- The advertised name may be truncated (section 2.2).

10. LEDs

The LED manager arbitrates between sources. Only the highest-priority active source is shown, except that the crash pattern alternates with one companion source every 1.5 s.

Priority (highest first): Factory test > **Crash** > Charging > Calibration > BLE event > Battery > BLE heartbeat > Activity.

Indication	Pattern	Colour
Activity heartbeat, active	100 ms on / 900 ms off, repeating	Teal (green + blue)
Activity heartbeat, inactive	100 ms on / 4900 ms off, repeating	Teal
Advertising	100 ms on / 900 ms off, repeating, until advertising ends	Blue
Connected	One 2 s flash on connect, then returns to heartbeat	Green
Crash / unacknowledged offline data	Red SOS-style blink, 250 ms on/off with pauses. Shown after a major crash and **at boot if offline records exist**. Cleared by acknowledge (`0x1002` = `0x01`) or by deleting offline data (`0x6166` = `0x02`).	Red
Battery status (when battery notification is sent)	500 ms on / 500 ms off: 3 blinks ≥ 90 %, 2 blinks 31–89 %, 1 blink ≤ 30 %	Blue
Charging, below full	Blue PWM ramp animation	Blue
Charging, full (≥ 4.19 V)	Solid	Blue
Calibration	Blue blink 250 ms on / 100 ms off while calibrating; one 2 s blue flash when done	Blue
BLE-set colour (testing only)	Solid, until cleared	As commanded (R, G, B, each 0 to 100)

When the crash LED is active, the companion (heartbeat, battery or BLE) alternates with it so the helmet still shows it is alive.

Because the red and green LED pins are swapped on some hardware revisions, verify colours against the physical helmet.

11. Typical flows

11.1 First connect after double tap

1. User double-taps the helmet (helmet not connected). Helmet starts advertising (blue blink, 20 s window).
2. App scans, identifies the helmet by manufacturer data (company `0x0ED6`, bytes `08 08 04 B3`) and reads battery % and record count from the advertisement.
3. App connects. Helmet flashes green for 2 s.
4. App requests MTU ≥ 129, discovers services, enables notifications on `0x6166` and `0x1002`.
5. App writes the RTC (`0x1003`).
6. App may send `0x0A` to `0x1002` for an IMU timestamp (`TS:...`).
7. If `record_count > 0`: app writes `0x01` to `0x6166`, receives 64 packets per record and the end marker `01 33 55 AA`.
8. When the app has stored the data, it writes `0x02` to `0x6166` to delete it. The red crash LED clears.
9. App disconnects. The helmet does not advertise again until the next double tap.

11.2 Crash while connected

1. Helmet notifies `C:tick` / `MINOR CRASH` and classification text (`Impact`, `Fall` or `CRASH-FF`, then `ires:`).
2. Depending on `crash_mode` (section 9.2) the helmet either streams 64 packets on `0x6166` (online) or stores the crash to flash.
3. App acknowledges with `0x01` to `0x1002` to stop the red LED.

11.3 Crash while disconnected

1. Helmet captures the crash, stores it to flash with a crash ID and timestamp, shows the red LED, then starts advertising (20 s). The record count in the advertisement increases.
2. App (or user double-tap) connects later and follows 11.1 steps 3 to 8.

12. Known behaviours and open points

These are observations from reading the source. Items marked *verify* should be confirmed on hardware before being relied on.

#	Item	Impact / recommendation
1	**Advertising timeout is 20 s** in firmware, while the feature list asks for 30 s.	Documented as built (20 s). A firmware change is needed to reach 30 s.
2	**Advertised name is shortened** (`QuinPro080`) for the default 11-character name. *Verify with a scanner.*	Match by manufacturer data, not name.
3	**Timestamp is not in the BLE crash packets.** It is stored in the flash record but the 126-byte packet carries only the crash ID.	The app must timestamp on receipt, or the firmware must be extended if event time is required.
4	**Device clock increments every 1 ms** (named "seconds"). *Verify.*	Agree the unit before using timestamps.
5	**IMU extended timestamp is approximate** (overflow counted as `1<<24`; interrupt fires at ≈`0xFF0000`).	Error up to ~0.4 % over long periods.
6	**Behaviour after a mid-transfer disconnect** during an offline replay is not explicitly handled in the firmware. *Verify.*	Avoid disconnecting mid-transfer; re-request after reconnect.
7	**Battery string format.** The firmware builds the string in place with `snprintf` using the same buffer as source and destination. *Verify the exact text on hardware.*	Parse defensively: split on commas.
8	**Pre-event rate 104 Hz, post-event 52 Hz, high-g ≈1 kHz** come from the firmware configuration and comments. *Verify.*	Needed for time-axis reconstruction.

#	Item	Impact / recommendation
9	**No pairing / bonding**; all characteristics are open.	Consider as a security point if the helmet will be deployed widely.